MIL-C-496D
14 December 1982
SUPERSEDING
MIL-C-496C
28 June 1968

MILITARY SPECIFICATION

CLIPS, END, STRAP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for end clips used on textile straps.

1.2 Classification. Clips shall be of the following types, classes, and sizes as specified (see 6.2 and 6.5).

Type I - MS 51926 (Ball)
  Class 1 - Brass
    Size - 5/8-inch, 3/4-inch, 1-inch, 1-1/4-inch, and 1-1/2-inch.
  Class 2 - Steel
    Size - 5/8-inch, 3/4-inch, 1-inch, and 1-1/4-inch.
Type II - MS 70109 (Flat)
  Class 1 - Brass
    Size - 2-inch.

* 2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5340

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SPECIFICATIONS

FEDERAL

QQ-B-613	-	Brass, Leaded and Nonleaded Flat Products (Plate Bar, Sheet and Strip).
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-E-480	-	Enamel, Baking, Phenol-or Urea-Formaldehyde.
MIL-F-495	-	Finish, Chemical, Black, for Copper Alloys.
DOD-P-16232	-	Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals).

STANDARDS

MILITARY

MIL-STD-105		Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads.
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment.
MS 51926	-	Clips, End, Strap - Type I.
MS 70109	-	Clips, End, Strap - Type II.

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A-109 - Steel, Carbon, Cold-Rolled Strip

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.3).

3.2 Materials. Materials shall conform to the requirements specified herein and in the subsidiary documents applicable thereto (see 6.6).

3.2.1 Brass. Brass sheet and strip shall conform to copper alloy numbers 268 or 260, quarter-hard temper of QQ-B-613.

* 3.2.2 Steel sheet and strip. Steel sheet and strip shall conform to cold rolled commercial quality and strip shall conform to cold rolled, temper No. 5 of ASTM A-109. Chemical composition of the steel shall be within the range of SAE numbers 1009 - 1020.

* 3.2.3 Enamel. Enamel shall be black conforming to type I or II of MIL-E-480.

3.3 Construction. Clips shall be constructed in accordance with the design, detail, and dimensions, as shown on applicable Military Standard and as specified herein. The type I clip shall be as shown on MS 51926. The type II shall be as shown on MS 70109 and formed with the smooth edge of the material on the outside of the clip (see 6.4). When clips are assembled to a strap and tested as specified in table II, no split edges shall be in evidence.

3.3.1 Type I clips.

3.3.1.1 Class 1. Type I, class 1 clips shall be fabricated of brass specified in 3.2.1 and as shown on MS 51926.

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3.3.1.2 Class 2. Type I, class 2 clips shall be fabricated of steel specified in 3.2.2 and as shown on MS 51926.

3.3.2 Type II Class 1 clips. Type II, class 1 clips shall be fabricated of brass specified in 3.2.1 and as shown on MS 70109.

3.4 Finish. The clips shall be finished as specified herein. Prior to finishing, all clips shall be thoroughly cleaned in accordance with commercial practice and shall be free of corrosion, dirt or discoloration.

3.4.1 Class 1, brass. Unless otherwise specified (see 6.2), type I and II brass clips shall be given a black chemical finish conforming to MIL-F-495 and tested as specified in table II. When clip is assembled to a strap and tested as specified in table II, the black chemical finish shall not flake.

* 3.4.2 Class 2, steel. Unless otherwise specified, class 2 clips shall be zinc plated and phosphate finished in accordance with type IV of ASTM B-633, followed by bake enameling. The enamel shall be as specified in 3.2.3. The enamel coating shall level out to a uniform, dry film thickness without orange peel, wrinkles, drops, streaks, or areas of no film. When clip is assembled to a strap and tested as specified in table II, the enameling on the clips shall be free of cracks or spalls.

* 3.4.2.1 Class 2, steel optional finishes. When specified (see 6.2), the following finishes on class 2 clips will be permitted.

(a) Phosphate coating conforming to type M or 4, class 1 or 2 of DOD-P-16232.
(b) Zinc plating conforming to type III, of ASTM B 633.

When clip is assembled to a strap and tested as specified in table II, the coating shall not flake, exposing bare metal.

3.5 Workmanship. The clips shall be free from digs or gouges, burrs, slivers, sharp edges (except that on the type II clip, rough or sharp edges shall be permitted on the inside edge), dents or tool marks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and tested as specified in 4.4.4. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

4.4.2 In-process inspection. In-process inspection shall be conducted to insure that clips are thoroughly cleaned and free of corrosion and dirt before finishing (see 3.4) and that zinc plating (see 3.4.2) has been accomplished. Whenever nonconformance is noted, corrections shall be made to the affected items and process.

4.4.3 End item inspection. The inspection lot shall consist of all completely finished specified clips of the applicable MS offered for inspection at one time. The sample unit for this inspection shall be one clip.

4.4.3.1 Visual examination. The clips shall be examined for the defects listed in table I. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, and 6.5 total (major and minor combined) defects.

TABLE I. Visual defects

Examine	Defect	Classification Major	Minor
Color and finish	Not finished or the type of finish is not as specified.	X	
	Dents, nicks, pores, abrasions, pits, or scratches.	X	
	Discoloration clearly noticeable.	X	
	Foreign matter imbedded in finish of surfaces exposed to view when in use.	X	

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TABLE I. Visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Color and finish (cont'd)	Foreign matter imbedded in finish, but not exposed to view when in use		X
	Coating is not level, uniform, and continuous, e.g. runs, orange peel, wrinkles, drops or streaks.	X	
	Area of no film.	X	
	Finish is tacky to touch	X	
	Plating (when applicable) is broken, cut through, corroded, or erupted.	X	
Type	Not as specified or as shown on Military Standard.	X	
	Type II clip not formed with smooth edge on outside.	X	
Construction and workmanship	Component cracked, fractured, malformed, distorted, or bent.	X	
	Digs, gouges, burrs, slivers, sharp edges, dents, or tool marks (rough or sharp edges on inside of type II clip permitted).	X	
	Edge not clean, e.g., sharp burr or fin.	X	

4.4.3.2 Dimensional examination. The clips shall be examined for compliance with dimensions specified. Any dimension not within the specified requirements shall be classified a defect. The inspection level shall be S-2 and the AQL shall be 4.0, defects per hundred units.

* 4.4.4 End item testing. The finished clips shall be tested as specified in table II for the characteristics shown therein. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units. The sample unit shall be three (3) clips.

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TABLE II. Test methods

Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as
Split edges showing	3.3	4.5.1	1	Pass or fail
Type I & II Black chemical finish for brass clips:				
a. Color & gloss	3.4.1	MIL-F-495	1	Pass or fail
b. Resistance to hot soap solution	3.4.1	MIL-F-495	1	Pass of fail
c. Resistance to accelerated weathering	3.4.1	MIL-F-495	1	Pass of fail
Integrity of enamel coating	3.4.2	4.5.1	1	Pass or fail
Integrity of optional finishes	3.4.2.1	4.5.1	1	Pass or fail

* 4.4.5 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with section 5 requirements. Defects shall be scored as specified in table III. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Lots submitted for examination shall be fully packaged except that containers shall not be palletized and need not be sealed or closed. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

TABLE III. Packaging defects

Examine	Defect
Marking (exterior location, and interior)	Omitted, incorrect, illegible, of improper size, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Bulged or distorted container.
Content	Interior containers more or less than specified. Quantity per box or container more or less than specified.

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4.6 Palletization examination. An examination shall be made to determine that palletization is in compliance with section 5 requirements. Defects shall be scored as specified in table IV. The sample unit shall be one palletized unit load, fully packaged. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

TABLE IV. Palletization defects

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limit.
Marking	Omitted, incorrect; illegible, of improper size, location, sequence, or method of application.

4.5 Methods of inspection.

4.5.1 Setting test for adequacy of finish and split edges. End clips shall be firmly set by any commercial method on a corresponding size strap and examined for compliance with requirements specified in 3.3, 3.4.1, 3.4.2, and 3.4.2.1 as applicable.

* 5 PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Clips of one type, class and size only, shall be packed in quantities as specified in table V in a snug-fitting intermediate fiberboard box conforming to grade W5c, style optional of PPP-B-636. Each box shall be tested from time to time to assure a compact and well-filled box. Each box shall be closed in accordance with the appendix of PPP-B-636.

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TABLE V. Quantity per intermediate packs and shipping containers

Type, class, size (inches)	Quantity per fiberboard box	Quantity per shipping container
Type I - (MS 51926)		
Class 1 and 2 (as applicable)		
Size 5/8-inch	3,600	21,600
3/4-inch	2,400	14,400
1-inch	1,700	10,200
1-1/4-inches	1,200	7,200
1-1/2-inches	1,000	6,000
Type II - (MS 70109)		
Class 1	900	5,400

5.1.1.2 Commercial. Clips shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Clips of one type, class, and size only, preserved as specified in 5.1, shall be packed in a wood cleated plywood box conforming to PPP-B-601, overseas type, style optional. The box shall be closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Unless otherwise specified (see 6.2), clips of one type, class, and size only, preserved as specified in 5.1, shall be packed in the quantity specified in table IV in a snug-fitting fiberboard shipping container conforming to style FTC, type CF, variety DW, class domestic, grade 350 of PPP-B-636. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

5.2.3 Commercial packing. Clips preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Palletization. When specified (see 6.2), clips packed as specified shall be palletized in accordance with load type I of MIL-STD-147, except that for overseas shipment the over-all height of the load shall not exceed 43 inches. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the container patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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5.4 Marking. In addition to any special marking required by the contract or purchase order, intermediate packs and shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. Clips are intended to keep strap ends from unraveling on clothing and equipage items.

6.2 Ordering data. Acquisition documents should specify the following

(a) Title number and date of this document and applicable MS.
(b) Type, class, and size required (see 1.2 and applicable MS), including applicable dash number and suffix.
(c) When a first article sample is required (see 3.1, 4.3, and 6.3).
(d) When finish other than black chemical is required (see 3.4.1).
(e) When optional finish is permitted (see 3.4.2.1).
(f) Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
(g) When palletization is required (see 5.3).

* 6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Sharp edges. Punching of the blank for the clip produces a typical sheared edge which is smooth on one side and rough or sharp on the opposite side (see 3.3).

6.5 Sizes. The clips are available only for webbing widths as shown on MS 70096 and MS 70109.

* 6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.7 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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